

Designing Targeted Explanations for Heterogeneous Stakeholders - AI for Collaborative Use in the Railways

Lena Schneider

Work and Organizational Psychology
ETH

Zurich, Switzerland
schneiderl@ethz.ch

Daniel Boos

Swiss Federal Railways
Bern, Switzerland
daniel.boos@sbb.ch

Prof. Dr. Gudela Grote

Work and Organizational Psychology
ETH

Zurich, Switzerland
ggrote@ethz.ch

Yaren Cakir

FH Graubünden
Chur, Switzerland
cakir.yaren@outlook.de

Abstract

The opaqueness of AI continues to represent a significant challenge when designing safe and reliable high risk-AI-systems, particularly for collaborative use by multiple heterogeneous stakeholders. Explainability approaches are used to address this opacity by increasing transparency and interpretability of AI-based systems and thereby establishing human control over the AI-system. In order to be effective, these explainability approaches need to be targeted to needs of all involved stakeholders. We show with an example use case from the railway sector how work design can be combined with technology design for successful human-system integration. Through observation of work processes and interviews with stakeholders at an early stage of system development we can support the design of targeted explanations for high-risk AI systems used in complex collaborative settings by gaining in depth understanding of the respective work systems. We identified five relevant explainability approaches for a visual train inspection decision support system. Exploratory testing then showed preferences for some explanations over others, fostering the discussion on choosing appropriate and effective explanations for visual tasks and informing the design of a final visual train inspection interface prototype for future research and application in the railways.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods; Field studies.

Keywords

XAI, Explainability, Accountability, Control, AI Ethics, Stakeholder Networks, AI in high-risk domains, Socio- technical systems design, Human-in-the-loop, Work Design

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1 Introduction

AI increasingly becomes part of work systems around the world. Its black-box characteristic leads to new challenges for socio-technical systems design that go beyond those known from previous technological innovations. Overcoming the opacity of AI-based systems to fully benefit from them [5] is even more relevant in safety-critical domains like the railway sector with strict regulatory oversight, where small failures can lead to disruptions or accidents with potential fatalities (e.g., malfunctioning of components leading to derailling of a train).

Article 14 of the EU AI Act [24] calls for a design and development of high risk-AI-based systems (including interfaces) that enable human oversight in order to minimize safety risks. This shall be achieved by enabling human actors to understand the capacities and limitations of the AI-based system and correctly interpret the system's output, to override or reverse the system output, to intervene or interrupt the system at any time while remaining aware of issues of over-reliance [24]. However, the application of this article presents a significant challenge for the development and deployment of AI-based systems in high-risk environments as it is still unclear how exactly the outlined system characteristics can be achieved.

We argue that creating targeted explanations for a system represents a central step towards fulfilling the above listed requirements of the EU AI Act as it supports the establishment of supervisory control of a human actor over an AI-based system by increasing transparency and interpretability and thereby enables the fulfilment of accountabilities. To design these targeted explanations requires gaining in depth knowledge about all human actors involved with the development, implementation and use of a high-risk AI system at an early stage of system development. In a work context, this knowledge includes understanding work processes and tasks as well as the collaboration amongst human actors. However, to

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successfully combine work design and technology development continues to present a significant challenge [16, 20].

The goal of the present late-breaking work is therefore to present insights based on a complex high-risk use case from the railways into how we can address at an early stage of system development the significant challenge of designing an explainable high-risk AI system for successful collaboration of multiple heterogeneous human actors in a complex setting with high task interdependency by drawing from work design methods. We do so by showing how gaining an understanding of the work system and processes for visual inspection of trains for damages as a first step can inform the design of a system interface with targeted explanations. First, we identified all human actors involved in the real-life use case through interviews and observations. Secondly, we analyzed the work processes and tasks as well as collaboration between stakeholders and collected first ideas for the design of relevant explanations, resulting in the inclusion of five different explainability approaches. Third, these approaches were then tested with a non-expert sample to explore potential general preferences for certain explanations over others in a visual task. Based on these findings, we created a first interactive prototype of an interface for the AI-based decision support system for visual train inspection for damage detection which serves as a base for a further application-grounded evaluation [8] with the real-life domain experts and informs the creation of the final system and interface.

2 Background

The demand for human oversight as stated in the EU AI Act [24] clearly places accountability with human users of high-risk AI systems but remains vague as to how these accountabilities can be fulfilled and which human actors should be held accountable to what end. Accountability very broadly refers to a human actor giving account towards human actors, where the actor is obligated to explain and justify their actions (see for example [1, 4]). In human control theories, human actors can only be held accountable for work processes they have control over [3]. Control can be described as the ability to predict, influence and understand processes and conditions [2]. To establish sufficient control for human actors is challenging [13] and can be attempted in two ways: through explainability approaches by increasing transparency and interpretability of AI-based systems for human actors and through function allocation between human actors and technology.

Explainability approaches are often applied to increase transparency and interpretability for one of two homogeneous main target groups: developers or end users [1, 18, 22]. However, the introduction of AI usually leads to highly intertwined “socio-technical landscapes” [21] and researchers have called for explanations to take into account the needs of different stakeholders [17] and the purpose the explanation aims to fulfil [14, 23]. In real-life work settings, these multiple stakeholder groups often have heterogeneous backgrounds (i.e., profession, education level, language, tasks). As these stakeholders not only interact with the system but also collaborate with each other, knowledge is also required on what this collaboration looks like and how stakeholders depend on each other throughout the different phases of development, implementation and use of new AI-based systems. As the previously clear distinction between technology development and use is much blurrier

for AI than for previous technologies [21] this poses an additional aspect to be considered when designing explanations.

The choice of effective explanations is not a simple one. Studies show that providing explanations does not always improve decision quality (see for example [10]), and even popular XAI methods like LIME, SHAP or TreeInterpreter can lead to a decrease in performance rather than a performance gain compared to when no explanations were provided [15]. Wang et al. [23] recently created a guide to choosing explanations based on the target group, phase in the system lifecycle and purpose of the explanation respectively. While this provides an important cornerstone for the design of explanations for stakeholders, we argue that for AI systems to be used in complex, collaborative settings in a safety-critical context, these criteria for the design of explanations alone are not sufficient and additional knowledge about the stakeholders and work system need to be attained and function allocation between human actors and AI system needs to be considered.

Function allocation refers to the distribution of tasks between human actors and technologies [7] and aims to optimize the performance of the system by combining the strengths of both humans and technical systems in a complementary way. Challenger et al. [6] underline that function allocation should be kept in mind by stakeholders involved in design, implementation and use of complex work systems and decisions about function allocation should be made early into the development process while involving different stakeholders through an iterative process. Grote et al. [13] furthermore suggest that the function allocation should be re-assessed regularly throughout the process of system development, implementation and use. By analyzing work processes and tasks, we can identify changes in function allocation introduced through the implementation of a new AI-based system and thereby address issues in regards to supervisory control when they occur. Function allocation additionally serves as the foundation to discuss accountabilities amongst stakeholders. The alignment of accountability and control in socio-technical systems is thereby seen as a central design principle [13]. As the introduction of opaque AI-systems leads to a shift of control capabilities from the human actors towards the system itself, the accountability-control gap known from traditional automation is widened [11]. Consequently, a need for re-negotiation of accountabilities based on the level of control for each involved human actor arises [2, 13].

3 Visual train inspection for damage detection: A high-risk AI system for a complex collaborative setting in the railways

In the present study, we focus on the use case of an AI-based decision support system for visual train inspection which will support human actors in detecting damages on train parts. The goal of introducing AI for visual train inspection is to allow for targeted assessments of wagons in case of potential detected damages, thereby reducing the financial cost of maintenance, increasing safety and punctuality of trains, and improving the efficiency of maintenance processes by optimizing the scheduling of maintenance appointments through condition monitoring via automated technical controls (including temperature and vibration sensors and cameras). AI use in the area of visual train inspection therefore has high potential to facilitate inspection processes.

Currently, the technical safety of trains and wagons is inspected on site by technical inspectors. This time-consuming, physically demanding inspection process also poses a potential health and safety risk to workers (for example as damaged train parts or insufficiently secured wagon loads can lead to accidents with severe consequences, including fatalities). This existing inspection process is therefore planned to be replaced. In the future, visual train inspections will be completed by a new AI-based system which will detect damages on images taken of the trains and wagons during normal operation by cameras stationed around the tracks. The system will then alert a human control center expert on any identified damages and the control center expert will decide over the prevalence of a damage and initiate measures with other stakeholders for repair of the damage based on system output. The introduction of this new AI-based system will lead to a significant change in function allocation without reducing the complexity or collaborative nature of the use case with high task interdependency amongst stakeholders. Not only will damage detection now be taken over by the AI-system, the same human actors who were before accountable for ensuring all damages were detected and addressed in a timely manner will now work in a control center where their key work task will be to oversee an opaque AI-based system rather than complete the inspection themselves. Furthermore, these actors will be deciding based on the system output which measures to take but will risk not being able to address prevalent damages due to the system failing to identify them (as false negatives cannot be identified by the control center expert). In addition to these changes in function allocation between the main target user group of the new system, the inspectors and the system itself, the introduction of the new AI-based system will widely affect the collaboration between the inspectors and multiple other stakeholders directly or indirectly (i.e., dispatchers, maintenance coordinators, maintenance technicians, conductors). An overview of these changes in function allocation and collaboration between all stakeholders as well as accountabilities is provided in Table 1.

4 Deriving and testing explanations

We used a multi-method approach to derive explanations for the visual train inspection AI-system. First we collected ideas for relevant explanations through interviews with the project team and document analysis as well as observations of the work processes for technical train inspections. Next, we tested these derived explanations following a human grounded approach [8] with an online sample of non-experts which completed a visual decision making task based on real-life task from the railway use case.

4.1 Interviews, Observation and Document Analysis

We conducted semi-structured interviews with different stakeholders of the visual train inspection use case. Interviews were conducted online using MS Teams or in person and were audio-recorded. Participation was voluntary and participants provided informed written consent. The interviews were approved by the ethics board of the university the researchers are employed by. In addition, we analyzed internal project documentation outlining the concept for the future AI system as well as the involvement of

different human actors. We furthermore observed different stakeholders during their daily work to analyze the underlying work processes and tasks and learn more about the collaboration between them. The safety expert within the project team asked for confidence indications as one explainability approach, while the project manager saw saliency maps as an explanation potentially helpful for developers. During observations we noticed the important role of the official error catalog for the visual train inspection process, which serves as a base for all damage protocols created by technical inspectors and guides all measures taken based on the type of identified damage. A maintenance worker furthermore expressed the wish for a text description rather than a visual explanation. As the project team wished for additional visual explanations, we included two additional explanatory approaches based on a literature review. We then created an initial interactive interface prototype encompassing all five explanatory approaches.

4.2 Online Testing

Participants

To gain insights into whether the chosen explanations were generally seen as helpful and appropriate in the context of the concrete railway decision-making task, we tested them online using Qualtrics. We recruited 109 participants via Prolific. All participants had completed a tertiary degree in engineering, computer science or a related field, but had no domain expertise of the railways. The real-life stakeholders for this use case vary widely in terms of professional background and education level (as can be seen in Table 1), we aimed to first achieve a basic understanding of how individuals with technical knowledge but no domain knowledge perceived the explanatory approaches. Compensation was 4 USD. We provided information on data processing, storage and handling and obtained informed consent. Participation was voluntary, and participants were free to withdraw from the study at any point in time.

Design

A within-subjects design was employed whereby we manipulated the type of provided explanation and measured performance. There were six experimental conditions representing the type of the visual or textual explanation, namely counterfactual explanations, text descriptions, confidence indications, saliency maps, integrated gradients, and no explanations. Explanations were derived based on interviews, observations and document analysis as outlined above. For counterfactuals as well as text descriptions, we used content from the official error catalog for the inspection of trains, freight, passenger, and dangerous goods wagons. To simulate the visual explanations, images were edited using Adobe Photoshop.

Material

Image Material. For the visual task, we used image material of different parts of trains and wagons showing different types of damages and wear and tear (i.e., broken wheels, damaged train doors). To ensure the task could be completed by non-domain experts, two student assistants first sorted the image material according to three categories, namely low, medium, and high difficulty to detect damage, respectively. For each task block, two images of the low difficulty, two images of the medium difficulty and one image of the high difficulty category were used.

Table 1: Overview of stakeholders involved in the use of a new visual train inspection decision support AI system

<i>Stakeholder roles</i>	Technical Inspector	Maintenance Dispatcher	Workshop Coordinator	Mobile Repairs Team	Shunter
<i>Before the introduction of the new AI-based system</i>					
<i>Key work tasks, function allocation</i>	Completes technical train inspection / damage detection as well as recording damage and creating protocols Marks damaged wagons and completes small interventions Informs other stakeholders about damages and ensures that operational data matches train composition, load, and destination Held accountable for detecting all damages in timely manner and initiating measures for repairs	Receives information about damages. Schedules repairs based on information provided by the technical inspector and capacity of workshops Informs workshop coordinator about upcoming repairs	Receives and shares information about scheduled repairs Assigns and supervises technicians for repairs Documents and checks completed repairs Provides operational release and communicates with maintenance dispatcher Relies on information in the damage protocol Held accountable for correct operational release	Repairs damaged wagons which cannot be transported to workshop on site Relies on damage protocols created by technical inspectors and schedule created by coordinator in collaboration with maintenance dispatcher to prepare and bring correct materials and tools and repair damages Accountable for successful repair which enables train to be transported to workshop	Removes damaged wagon from train composition and moves them to siding to wait for transport to workshop or on site repairs Moves trains around to operating points Relies on information by technical inspector, and maintenance dispatcher on which wagons have to be moved Couples wagons
<i>After the introduction of the new AI-based system</i>					
<i>Key work tasks/ Function Allocation</i>	AI takes over task of damage detection and protocol generation Technical inspectors take on the new role of control center expert, assess system output and confirm or revoke suggested measures Inform other stakeholders Will still be held accountable but have reduced control as they does not work on site anymore and cannot view damages with their own eyes and have no control over how the algorithm processes images and comes up with suggestions (leading to an issue of potential false negatives)	Continues to rely on information about damages via damage protocol generated by AI and confirmed by control center expert to schedule repairs Can access AI-based system to review model suggestions and collect additional info on train to assist scheduling	Schedules repairs based on damage protocol and additional information from maintenance dispatcher, technical inspector and AI-system Requires clear explanation by the system of where damage is located and what it entails to coordinate repairs for technicians. Documentation Provides operational release	Can verify and assess damage with the help of the new system in order to prepare correct materials and tools for repairs. Still receives schedule and additional information from maintenance dispatcher and technical inspectors about upcoming repairs	In addition to previous work tasks shunters will now be assigned the tasks of completing targeted checks of identified damages, conducting small interventions, confirming operational data or marking damaged trains Further training will be needed to gain knowledge and skills for taking on the additional tasks

Additional stakeholders indirectly being affected by the introduction of the new AI-based system will be shift managers and maintenance technicians as they work within the same work system and collaborate with the stakeholders listed above, but not the AI-System.

Task and Procedure. Participants were told they would interact with an AI-based decision support tool which assists them in completing a visual decision-making task from the railways by detecting damages on images of train parts and providing additional explanations for the alert. After completion of all blocks, participants were debriefed. Within the binary choice task, participants were asked to decide based on a provided black and white image whether there was a damage on the presented train part or not. Images were shown across six blocks, each consisting of five visual tasks, respectively. Participants started with a block of five tasks where images were provided without any additional explanation (i.e., no explanation condition). Each of the following five blocks showed a specific type of explanation in addition to the image material, so that the type of explanation within each block stayed the same. The order of blocks after the first block was randomized. Upon task completion, participants were asked to state the most helpful out of all explanations provided and to elaborate on their choice (open text box). Additionally, participants were asked what additional explanations they would like to be added and what else they wished for in terms of design (open text box).

Perceptions of the Chosen Explanatory Approaches.

44.1% of the participants saw text descriptions as the most useful explanations. What participants particularly liked about text descriptions was how clear, straightforward and easy to understand they were, the detailed information on the damage that they provided, and the help in identifying which parts of the image should be assessed by the human actor in order to confirm whether there was an underlying damage. The text descriptions furthermore included potential reasons for damages and options for measures to take, which was also seen as helpful and as leaving less room for misinterpretation. 24.5% of participants perceived the integrated gradients approach as most useful. Participants reported that the integrated gradients method was understandable without a deeper expert knowledge, was straightforward and outlined relevant areas clearly, furthermore it was seen as easy to follow and visually user-friendly. 16.7% of participants found counterfactuals to be the most useful explanation. Counterfactuals were seen as helpful because they allowed for a comparison between the presented image of the train part with potential damage and an intact (or otherwise damaged) train part. Counterfactuals were therefore seen as clear, quick, easy and efficient to use compared to the other explanations, that were also seen as requiring more time by one participant. One user reported that the counterfactuals made them feel like they allowed them to engage themselves and not fully rely on where the tool was telling them to look at. 13.7% of participants found saliency maps to be the most useful and saw them as easy to understand and simple. Participants liked that saliency maps provided a visual explanation, helped seeing which parts of the shown train part was most important, and one participant also stated that it was the only method that increased their trust in the AI. Only one person perceived confidence indications as the most useful explanation but did not provide further information.

When asked about what participants wished for in terms of additional explanations or design, 25 participants mentioned that they wished for an option to combine multiple or all explanations in one interface (either to choose from or all being displayed at once).

5 Discussion

We address the challenge of providing sufficient control for human actors as required per the EU AI Act[24] by outlining two paths to increased control: through designing targeted explanations addressing stakeholder characteristics and needs, and effective function allocation between human and technology.

AI-based systems have to support the completion of work tasks and collaboration of stakeholders. Understanding work processes therefore serves as a foundation for system development and extends our perspective on what knowledge about human actors is needed in order to design explainable AI based systems. We explored what human oversight means in the context of the visual train inspection use case and how we can support it through design. In a complex, collaborative high-risk setting, the introduction of a new AI-based system is going to lead to far-reaching consequences for function allocation and collaboration. While the role of the technical inspectors is going to change the most drastically, other stakeholders within the work system and their collaboration amongst themselves are also directly and indirectly affected, showing that all stakeholders need to be considered. Moreover, function allocation between the control center experts and the AI-based system as anticipated for this use case will lead to a disbalance between accountability and control for technical inspectors: As control center experts they will be tasked with overseeing the AI-based system and are accountable for identifying damages, but without increased transparency and interpretability through effective explainability approaches the control demands will not be fulfilled.

We used the insights gained through interviews and observations as a first step to create an interface prototype with different explanations. The following exploratory testing of these explanatory approaches showed clear preferences of participants for certain explanations: almost half of the participants perceived text descriptions as the most useful explanations for the visual task. Additionally, participants also expressed the wish to be provided with multiple explanations at once to be able to choose and combine several explanatory approaches. While existing research generally refers to choosing one explainability approach to be implemented in an interface, our findings indicate that there is no one size fits all approach to explanations and explanations may be combined effectively to enable an augmentation of human performance. Our example use case from the railways furthermore underlines the importance of purposeful work design. Involving all stakeholders in the development process of AI and considering their needs when designing AI-based systems thereby provides a way to reduce the risks stemming from the introduction of AI in complex, safety-critical contexts.

Next Steps

For the initial exploratory testing completed in the present study we relied on non-domain experts rather than domain experts. As the time of domain experts is very valuable and access for research purposes therefore limited, we were interested in identifying potential general preferences of some explanations over others first to serve as a base for further evaluation. However, we cannot exclude that the various domain experts will be different in their preference for certain explanations due to their vast domain knowledge and

heterogeneous background. For example, the confidence indications, which were asked for by the safety expert of the use case project team, were not seen as particularly useful in the initial testing. This is nevertheless in line with our assumption that different stakeholders will vary in their preferences and underlines the need for targeted explanations in line with work processes and tasks of key user groups. In regards to preferences by the real-life stakeholders, we assume that stakeholders with lower levels of expertise might benefit particularly from counterfactuals and text descriptions due to their perceived clarity to identify damages. Text descriptions furthermore indicate potential measures to take after a damage was detected, thereby assisting in addressing damages effectively. A preference for text descriptions was also expressed by the maintenance coordinator and maintenance dispatcher, while the project manager indicated that developers might mostly rely on integrated gradients or saliency maps to fulfil their work task of monitoring the algorithm.

Following the recommendation of Doshi-Velez et al. [8], further evaluation will be conducted using an application grounded approach: based on the present findings we will create a mature version of an interactive interface prototype with explanations for a follow up study with real-life stakeholders (including for example maintenance dispatchers, technical inspectors, or coordinators for maintenance and repairs) of the AI-based visual train inspection system. Using the Thinking Aloud method [9], participants will be asked to freely verbalize their thoughts while interacting with the system to provide insights into their thought processes and decision-making strategies. This will allow for an evaluation of the chosen explanations in terms of suitability and support the design of targeted explanations for specific user groups. A final, high fidelity prototype will then be created and recommendations for systems design will be derived.

6 Conclusion

The present late-breaking work explores how the significant challenge of designing an explainable high-risk AI system for successful collaboration of multiple heterogeneous human actors in a complex setting with high task interdependency can be addressed in an early stage of system development. By taking on a strong process focus, we showed how knowledge about stakeholders and their needs based on the analysis of work processes and interdependencies can serve as a base for designing explainable AI based systems and provided a method for initial testing of different explainability approaches for a specific high risk use case. We furthermore underline the role of designing work processes as a way to provide control to human actors that are being held accountable for the consequences of decisions made with the support of an AI-based system and thereby consider the design principle of aligning accountability and control.

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